



Machine Learning for Cybersecurity

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Introduction

- Problems ML can solve
- ML viewpoint
- Data makes a security problem solvable by ML
- Feature representations

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Cybersecurity Problems Tackled by ML



- Phishing detection
- Spam detection
- (D)DOS detection
- Botnet detection
- Intrusion detection system (IDS)
- Ransomware detection
- Malware detection
 - android
 - linux
 - windows
- Biometrics
 - fingerprint, eye, face detection
- Keystroke dynamic authentication
- Credit card fraud
- Anomalous user behavior
- Insider threat detection
- Penetration testing (is used in fuzzers)

ML tasks used for cyber security

- Detection:
 - phishing, spam, malware, ransomware, intrusion, biometrics
- Anomaly and Clustering:
 - fraud, insider threat, IDS, IPS
- Reinforcement Learning
 - fuzzing, IDS

Data Makes ML Work

- It is attractive to say “we will solve that with ML”
- What is required to create an ML solution for any given (cybersecurity) problem?
- Data!
- What kind of data would we collect for all the different security problems listed so far?

Phishing and Spam Features

- Phishing / Spam
 - textual features (similar to NLP)
 - n-grams
 - counting features
 - embedding vectors (deep learning)
 - domain specific
 - IP addresses, domain reputation
 - SSL certificates
 - age of website

Malware Features

- Header information
- Static features
 - assembly instructions
 - Android manifest
 - PE files raw bytes (entropy histograms)
 - structural PDF features
- Dynamic features collected
 - api / function calls made

Selecting features requires domain knowledge

- A strong understanding of the problem itself. What is the attack, how does it work.
- What is the data we can get which represents the attack?
- You need a way to extract a numerical representation of your data
- You also need to have some idea what would help the algorithms *learn* to differentiate between the classes

Conclusion

- Security tasks that ML is used for
- Different ML tasks that can be used for security
- Data is needed to make it work.
- Feature extraction is a tricky part

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Can ML Help?

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Introduction

- Who needs ML
- Understand the domain problem
- What questions should you ask
- Identify features and data

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Understand the Problem



- The first step is to understand the problem that needs to be solved.
- Example: malware detection
- There are people who write malicious software that is embedded in legitimate software
- They do it in such a way that nobody notices, yet when you run the legit software, the malicious software automatically runs in the background



Understand the Goal

- Ok, we understand that malware exists, but what would you like to do about it?
- Currently we can identify malware by manually examining the software using a number of tools.
- This is very time consuming, we would like to automate this with ML
- Question: "what would be the input to the tool?"
 - Answer: "The input would be a software program"
- Question: "What would be the output?"
 - Answer: "A decision as to whether the program is malicious or not"

Q: What kind of data do you have?

- A: We have lots of data!
- Q: Great, let's drill down a bit, what would a single sample look like. How is it stored. Also, let's quantify "a lot", how much is it really?
- Q: Is the data labeled? Do we have a list of programs that are labeled as benign or malicious? How many for each class?

Q: What does a domain expert do when they solve this?



- A: Well typically I would do this....
- At this point you need to do on a deep learning dive with your client
- The entire time you are learning about how to they solve their problem manually, you need to consider which features you may want to try and extract from the data.
- The goal is to understand what patterns the human is looking for, so you can automate it into the features.

Conclusion

- Everyone wants AI/ML, but it's not always practical (or wise)
- Understand the problem and goal (input / output)
- Understand the process
- What data is available
- Look for public datasets

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Example: Phishing Detection

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Introduction

- What is phishing
- Blacklists
- Steps for automations
- Features
- Considerations

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Phishing



- Phishing is a masquerading attack where the attacker pretends to be a legitimate website in order to steal the credentials from a user.
- Attacks will send *lures* which can be email, sms, any form of communication that contains a link.
- The goal is to direct users to a website that is fake, it is calling a "phishing website" and there the users are generally asked to "sign in" for some service.
- It would be your email, google account, facebook, bank account etc. When you sign in you are unknowingly giving your credentials to the attacker

Blacklists

- The easiest way to deal with phishing is detection.
- We would like to block users from ever visiting phishing websites.
- Traditionally this has been with black-lists, lists of known malicious websites.
- The problem is that a phishing websites can be put up and taken down very quickly
- Blacklists require manual effort, they go out of date very quickly

Automation with ML



- We would like to build a machine learning algorithm that accepts as input a URL and gives as output a label “phishing” or “benign”
- What is the kind of data need?
- We need a list of benign websites
- We need a list of phishing websites

Feature extraction

- What tells us if a website is a phishing websites?
- We closely examine the URL – if the page looks like a facebook sign in, does the URL contain facebook in it?
- Which URL is safe to click on?
 - <https://www.facebook.com>
 - <http://www.facebook.com>
 - <https://facebook.com/give-me-your-login>
 - <https://abc.nyz.faceb00k.com/sign-in>
- Therefore, the *domain* is the most important part of the url

URL Domain features

- We could have a whitelist of safe domains – and then a feature could be “is the domain whitelisted”?
 - However, this falls apart quickly, because many times phishers compromise legitimate webpages (with good domains) and inject phishing pages *into* the safe websites!
- Length of the domain
- Length of the URL
- Number of paths (/)
- Number of parameters (?)

HTML Features

- Is it asking you to login?
 - extract forms
- Are there logos placed?
- Number of images
- Amount of text in the website
 - statistically phishing websites have less text and source code than benign websites
- The number of javascript tags

WHOIS features

- Where is the site hosted (geographically)?
- How old is it?
- Any identifying information?



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Lookup features

- Attempt to automate the *target* of the site (who is the site supposed to be?)
- Automatically search for the site on google and take the top match
- How similar are the pages?
- Use this as a feature

Considerations

- How long will your model take to make a decision?
- Certain features are much easier to capture than others
- How much confidence will your model have?
- Which metric should you optimize?
 - What is the tradeoff between precision and recall?
- Where do you get the labeled data?
- Which classification models should you use?

Conclusion

- Looked at phishing in depth
- Brainstormed many features
- Considered tradeoffs for different situations

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Public Datasets

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Introduction

- Public data
- How was it collected
- Reputable?
- Usable
- Examples
 - UNB
 - UCI
 - Kaggle

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Datasets

- There are a *lot* of public datasets you can get
- I will list them out in the resources
- What are the questions you should ask when you look for public datasets?

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How was the data collected?

- What is the methodology they followed when they collected the data?
- What makes you think you can trust their *ground truth* or not?
- Does their reporting / write up convince you that it's well done?

Is the source reputable?

- Quality of the website
- Reputation of the source
- Do other people use it (and who are they)



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Is it easy to use?

- How large is it?
- What format does it come in?
- How much work would you have to do?



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UCI



UNB



Kaggle



Conclusion

- How to examine a public dataset
- Where and how was it collected?
- Is it reputable
- Can you use it easily?
- Looked at a few live

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